Dog Breed Classification with Mixed Detection

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Repo:`https://github.com/AashleshaKhanna/DogBreedClassifierWithMixedDetection`

Introduction

Dog breed classifiers that perform well on curated datasets often fail in real-world photos (motion blur, poor lighting, partial views, cluttered backgrounds) and may output *overconfident* incorrect breeds. This becomes especially problematic for **non-dog images** and **mixed-breed dogs**, which are common in practice but do not match "pure-breed" training labels. My goal is a practical system that takes a casual photo and outputs either (a) a breed prediction with **reliable confidence**, or (b) **"unknown"** when the input is ambiguous or out-of-distribution (OOD). This improves trustworthiness for pet owners, shelters, and veterinary settings where incorrect labels can mislead downstream decisions. Deep learning is appropriate because breed recognition is a fine-grained vision task requiring robust, learned visual features under significant appearance variation, and modern methods support transfer learning, calibrated confidence, and OOD rejection.

Illustration

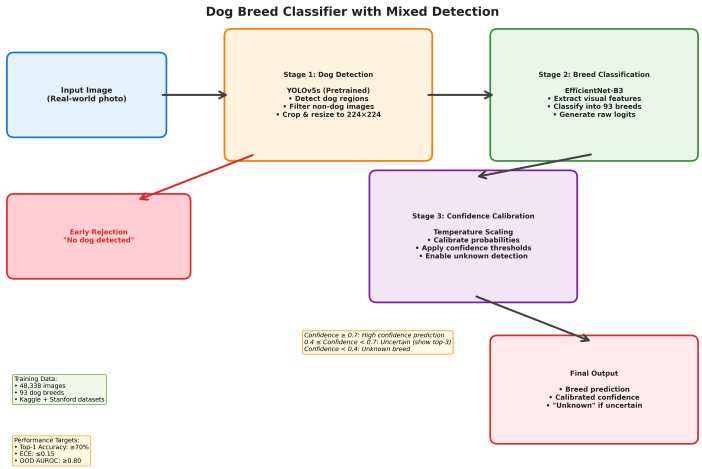

Figure 1: Pipeline overview: (1) dog detection + crop (reject non-dog), (2) fine-grained breed classification over 93 breeds, (3) calibrated confidence + thresholding to output "unknown" for mixed / ambiguous / OOD inputs.

Background & Related Work

- **Fine-grained dog breeds:** Stanford Dogs established breed classification as a fine-grained benchmark and motivates localization to reduce background bias (1).
- **Robustness evaluation:** ImageNet-C shows standard CNNs degrade under common corruptions (blur, noise, lighting), motivating explicit robustness testing (2).
- **Calibration:** Modern neural nets are often miscalibrated; temperature scaling is a simple and strong post-hoc calibration method (3).
- **Confidence-based OOD:** Baselines using max softmax probability quantify uncertainty for OOD/misclassification detection (4).

- **Open-set recognition:** OpenMax explicitly models an "unknown" category for deep networks, aligning with our rejection goal (5).

Data Processing

I will combine two dog-breed datasets with different image styles: (i) *Kaggle Dog Breed Classification* (6) and (ii) *Stanford Dogs* (1). I will restrict to the **93 overlapping breeds** and create a canonical label list. All breed strings will be **normalized**. I will (1) drop unreadable/corrupt images (verified by loading), (2) remove duplicates using file-hash matching, and (3) convert images to RGB. Kaggle will define the held-out **validation/test** benchmark; Stanford images will be used to **augment training** only for the 93 overlapping breeds to reduce overfitting to one collection style. Any additional splits I create will be stratified by breed with a fixed random seed. Inputs will be **resized** to the backbone resolution (300–380 px), normalized with ImageNet mean/std. Training augmentations such as random resized crop, horizontal flip, and mild color jitter will be done. Test-time uses deterministic resizing/center-crop. From the clean test split, I will generate **corrupted variants** (Gaussian blur, brightness/contrast reduction, random occlusion/crops) following common-corruption evaluation practices (2). I will collect non-dog images from ImageNet and COCO (12; 8) to evaluate **rejection** performance.

Architecture

Stage 1 (Dog Detection): A COCO-pretrained YOLO detector will identify dog bounding boxes and crop the dog region; images without a confident dog detection will be labeled "unknown" (9). Cropping will reduce background reliance and improve fine-grained focus. **Stage 2 (Breed Classification):** A transfer-learned CNN (EfficientNet family) will predict one of 93 breeds from cropped regions (10). I will fine-tune with cross-entropy and standard augmentation. **Stage 3 (Confidence Calibration):** Temperature scaling will calibrate the classifier logits on validation data (3).

Baseline Model

I will train a classical ML classifier on fixed image features. For each image, I'll first use the Stage-1 dog detector to crop the dog region, then resize to 224×224 and extract a 2048-D feature vector from an ImageNet-pretrained ResNet-50(11) by taking the global average output of the last convolutional layer. I will then train a one-vs-rest linear SVM to classify among the 93 breeds using these frozen features. Hyperparameters will be selected on the validation set. This baseline is simple, fast to reproduce, and provides a strong non-neural decision layer to compare against our calibrated pipeline.

Evaluation & Success Metrics:

- **Breed classification (clean):** Top-1 and Top-5 accuracy on a held-out clean test set (standard in large-scale image classification) (12).
- **Robustness:** Accuracy on corruption test sets (e.g., blur, low-light, occlusion) and the relative drop vs. clean accuracy, following common-corruption evaluation protocols (2).
- **Calibration:** Expected Calibration Error (ECE) before vs. after temperature scaling, plus reliability diagrams on the validation set (3).

Success criteria: Compared to the baseline, we aim to (i) achieve comparable clean Top-1/Top-5 accuracy, (ii) reduce corruption-induced degradation, and (iii) reduce ECE after calibration.

Ethical Considerations

Automated breed prediction can be misused to support breed-restrictive policies; erroneous labels may harm owners and dogs. Training data may underrepresent mixed breeds or certain regions, potentially biasing performance. To mitigate, I will: (1) emphasize calibrated confidence and "unknown" outputs, (2) document limitations and recommend human verification for high-stakes use, and (3) analyze failure cases, especially mixed breeds and OOD inputs.

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